

MS COMPREHENSIVE EXAM FALL 2019 SOLUTIONS

Problem 1. Let $X_1, \dots, X_n$ be i.i.d. with pdf f_θ where $\theta \in \{0, 1\}$ is unknown, and $f_0(x) \equiv \mathbf{1}_{0 < x < 1}$, $f_1(x) \equiv \frac{1}{\sqrt{2x}} \mathbf{1}_{0 < x < 1}$. Find the MLE of θ .

Solution 1. Note that

$$L(\theta = 0|\mathbf{X}) = 1, \quad L(\theta = 1|\mathbf{X}) = \prod_{i=1}^n \frac{1}{\sqrt{2X_i}}.$$

Therefore, the MLE is given by

$$\hat{\theta}_n = \mathbf{1}_{\prod_{i=1}^n \frac{1}{\sqrt{2X_i}} \geq 1}.$$

Problem 2. A random sample $X_1, \dots, X_n$ is drawn from a Pareto population with pdf

$$f(x|\theta, \nu) = \frac{\theta \nu^\theta}{x^{\theta+1}} \mathbf{1}_{[\nu, \infty)}, \quad \theta > 0, \nu > 0.$$

Find the MLEs of θ and ν .

Solution 2. The log likelihood function is

$$L(\theta, \nu|\mathbf{X}) = \left(n \log \theta + n \log \nu - (\theta + 1) \log \left(\prod_{i=1}^n X_i \right) \right) \mathbf{1}_{X_{(1)} \geq \nu}$$

For fixed θ , $L(\theta, \cdot|\mathbf{X})$ is increasing in $[0, X_{(1)}]$. So the MLE of ν is $X_{(1)}$, i.e. $\hat{\nu}_n = X_{(1)}$. On the other hand,

$$\begin{aligned} \frac{d}{d\theta} L(\theta, X_{(1)}|\mathbf{X}) &= \frac{n}{\theta} + n \log X_{(1)} - \log \left(\prod_{i=1}^n X_i \right) \\ &= \frac{n}{\theta} - \log \frac{\prod_{i=1}^n X_i}{(\min_{1 \leq i \leq n} X_i)^n} = \frac{n}{\theta} - T(\mathbf{X}). \end{aligned}$$

Solving for zero of the above equation yields that

$$\hat{\theta}_n = \frac{n}{T(\mathbf{X})}.$$

It is easy to check that $\hat{\theta}_n$ is a maximum so is the MLE.

Problem 3 A college recruiter wants to show that 10 years after graduation, the incomes of graduates from RU are higher than the income of those from University of Connecticut (UC) and of Penn State (PS). Five former students from each school were selected at random and their current salaries are given below in units of \$1000.

	<i>RU</i>	<i>PS</i>	<i>UC</i>
1	31	30	22
2	56	54	21
3	143	37	24
4	78	26	65
5	210	70	36

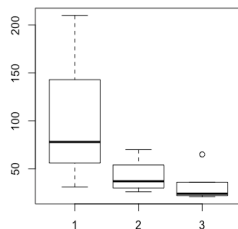
We are interested in comparing the salaries among the 3 samples (RU, UC, PS).

1. Draw a boxplot of the groups by calculating median, quartiles and extremes of the three samples. The p-value of the ANOVA F-test for comparing the means of the 3 groups is not significant (p-value=0.06). Can you interpret the boxplot and explain this p-value? Are the samples skewed? Do the samples meet the F-test assumptions?
2. Use the median and quartiles to construct a Spread-Vs-Level plot. From the Spread-Vs-Level plot determine if there is a relationship between spread and level and find the optimal power transformation that will eliminate it.
3. Perform an ANOVA F-test with the data after transformation. Is there a statistically significant difference among the group salaries?

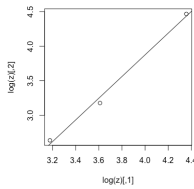
(Hint: The Spread-Vs-Level plot is performed by plotting the $\log(\text{Interquartile Range})$ vs $\log(\text{Median})$ and the slope b of this graph gives the suggested power transformation with power $1-b$.)

Problem 3 Solution.

1. Data is skewed not normal, variances are not equal, the ANOVA F-test is not appropriate. The boxplot shows big difference in favor of RU but the F-test p-value cannot detect it.



2. The Spread Vs level plot has a slope between 1 and 1.5 so the transformation is powers 0 to -0.5.



3. After transformation the ANOVA F-test is significant with p-value 0.04, which suggest that there is a difference between the salaries of graduates of the three schools after 10 years.

Problem 4 Suppose that the vector (X, Z) has a bivariate distribution given by:

$$f(x, z) = \begin{cases} 2e^{-(x+z)} & 0 < x < z < \infty \\ 0 & \text{Otherwise} \end{cases}$$

1. Give the probability distribution of random variable $Z|X=x$
2. Calculate the $E(Z|X=x)$

Problem 4 Solution

1. The marginal pdf of X is $f_x(x) = \int_x^{\infty} 2e^{-x-z} dz = 2e^{-x}$

Then $f_{Z|X}(z|x) = \frac{2e^{-(x+z)}}{2e^{-x}} = e^{x-z} \quad 0 < x < z < \infty$

2. $E(Z|X=x) = \int_x^{\infty} z e^{x-z} dz = x+1$

Problem 5. The below temperatures in Fahrenheit were observed in Piscataway and New Brunswick on the same date for both cities for 7 randomly chosen days in the month of July over the past 7 years (i.e. one day was randomly chosen in each year). For this question assume that the central tendency and variance of the temperatures in each city is normally distributed with a constant mean and variance for all days in July that is remaining the same over the past 7 years and into the future ... i.e. no global warming. But note that the means and standard deviations could differ between the two cities. You can also assume that measures taken in different years are independent but that measures taken the same day in Piscataway and New Brunswick are not independent.

Year	The Temperature Below the Year is for a Randomly Chosen Date in July Where the Date is the same for Both Piscataway and New Brunswick						
	2013	2014	2015	2016	2017	2018	2019
New Brunswick	78 °F	81 °F	90 °F	85 °F	88 °F	83 °F	83 °F
Piscataway	78 °F	79 °F	90 °F	82 °F	87 °F	82 °F	81 °F

A. Give separate point estimates for the average temperature for a random day in July and for the standard deviations of these estimates for New Brunswick and for Piscataway.

SOLUTION

New Brunswick

$$\text{Mean} = (78 + 81 + 90 + 85 + 88 + 83 + 83) / 7 = 84$$

$$\begin{aligned} \text{Variance} &= \frac{(78 - 84)^2 + (81 - 84)^2 + (90 - 84)^2 + (85 - 84)^2 + (88 - 84)^2 + (83 - 84)^2 + (83 - 84)^2}{7 - 1} \\ &= \frac{36 + 9 + 36 + 1 + 16 + 1 + 1}{6} = \frac{100}{6} = 16.67 \quad \text{STD-DEV} = \text{SQRT}(16.67) = 4.08 \end{aligned}$$

$$\text{Standard Error of the Mean} = 16.67 / \sqrt{7} = 1.54$$

Piscataway

$$\text{Mean} = (78 + 79 + 90 + 82 + 87 + 82 + 81) / 7 = 82.71$$

$$\begin{aligned} \text{Variance} &= \frac{(78 - 82.71)^2 + (79 - 82.71)^2 + (90 - 82.71)^2 + (82 - 82.71)^2 + (87 - 82.71)^2 + (82 - 82.71)^2 + (81 - 82.71)^2}{7 - 1} \\ &= \end{aligned}$$

$$\frac{22.18 + 13.76 + 53.14 + 0.50 + 18.40 + 0.50 + 2.92}{6} = \frac{111.4}{6} = 18.56$$

$$\text{STD-DEV} = \text{SQRT}(18.56) = 4.31$$

$$\text{Standard Error of the Mean} = 4.31 / \sqrt{7} = 1.63$$

B. Next year (2020) another day in July will be randomly chosen in New Brunswick and the temperature taken. Give a 95% confidence interval for what this temperature will be.

SOLUTION

$$84 \pm t_{6,0975} 4.08 \sqrt{1 + 1/6} \text{ or } 84 \pm 2.45 \times 4.46 \text{ or } (84 - 10.68 \text{ to } 84 + 10.68)$$

$$\text{or } (73.32 \text{ to } 94.68)$$

C. Test the null hypotheses that in July the average temperatures are the same for New Brunswick and Piscataway versus the alternative that are different with an overall Type 1 error of 0.05

SOLUTION

Temperatures on the same day are not independent for Piscataway and New Brunswick so look at paired differences

Day	1	2	3	4	5	6	7
New Brunswick	78	81	90	85	88	83	83
Piscataway	78	79	90	82	87	82	81
Difference	0	2	0	3	1	1	2

SOLUTION

The Mean paired difference is $(0 + 2 + 0 + 3 + 1 + 1 + 2) / 7 = 1.29$

The variance of a single difference is

$$\frac{(0 - 1.29)^2 + (2 - 1.29)^2 + (0 - 1.29)^2 + (3 - 1.29)^2 + (1 - 1.29)^2 + (1 - 1.29)^2 + (2 - 1.29)^2}{7 - 1}$$

$$= \frac{1.66 + 0.50 + 1.66 + 2.92 + 0.08 + 0.08 + 0.50}{6} = \frac{7.4}{6} = 1.23$$

So the standard deviation of a single difference is $\text{SQRT}(1.23) = 1.11$

And the standard deviation of the men difference is $\frac{1.11}{\sqrt{7}} = 0.42$

So the t-value is $1.29 / 0.42 = 3.07$ which is greater than $t_{6,0.975} = 2.45$ so reject H_0

Problem 6. Conservationists in France have captured 150 “dahuts”. However, they do not know the sex yet of each of these animals as such is hard to identify. It is known that on average 52% of dahuts are male. You can assume that the sex of each of the 150 captured dahuts is independent from that of the others.

A. What is the large sample estimate for probability that between 40% and 60% of the captured 150 dahuts are male?

SOLUTION

The expectation of the mean is 0.52

The standard error of the mean is $\sqrt{\frac{0.52 \times 0.48}{150}} = 0.0408$

So the large sample approximation to the probability to have between 0.4 and 0.6 dahuts being male is the probability that a standard normal variate falls between

$-2.94 = \frac{0.4 - 0.52}{0.0408}$ and $1.96 = \frac{0.6 - 0.52}{0.0408}$ which from standard normal probabilities is $0.975 - 0.002 = 0.973$

B. It costs on average 10 Euros to main a male dahut for one year. But this cost differs from bird to bird with a standard deviation of 3 Euros. Similarity it costs on average 8 Euros to main a female dahut for one year with again a bird to bird standard deviation of 3 Euros. What are i) the expected costs and ii) standard deviation of the expected costs to maintain a randomly chosen dahut (based on its probability to be male or female)?

SOLUTION

$E[X]$ is $0.52 \times 10 + 0.48 \times 8 = 9.04$

The variance is $E[X^2] - (E[X])^2$

Again $E[X]$ is $0.52 \times 10 + 0.48 \times 8 = 9.04$

So $(E[X])^2 = (9.04)^2 = 81.72$

$E[X^2] = 0.52 \times E[X^2] \mid \text{Male} + 0.48 E[X^2] \mid \text{Female}$

Now for each sex $E[X^2] = [\text{mean of that sex}]^2 + [\text{stddev of that sex}]^2$

Which is $10^2 + 3^2 = 109$ for males

$8^2 + 3^2 = 73$ for females

So the overall $E[X^2]$ is $0.52 \times 109 + 0.48 \times 73 = 56.68 + 35.04 = 91.72$

Thus overall $E[X^2] - (E[X])^2 = 91.72 - 81.72 = 10$

And the standard deviation is $\text{SQRT}(10) = 3.16$